

Health360 AI Customer Query and Complaint Deflection

Product Requirements Document (PRD)

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1. Executive Summary

Health360 receives a growing volume of customer conversations across product questions, order support, delivery concerns, returns, and complaints. The support team has limited human capacity, so repetitive questions slow down response times and create inconsistent answers. This is especially risky for a health supplements business because product guidance must be accurate, controlled, and compliant.

The proposed product is an AI-powered customer support layer on the Health360 website, backed by Health360 product data, policy content, complaint routing rules, and an admin dashboard. The system should deflect routine queries, classify conversations, provide safe product information, escalate uncertain or high-risk cases, and give administrators metrics for operations and continuous improvement.

2. Business Problem

Problem statement: Health360's support team cannot efficiently handle the rising volume of repetitive product, order, and complaint queries while maintaining fast SLAs, consistent answers, and compliant supplement communication.

Key Pain Points

- High query volume exceeds available support capacity.
- Customers repeat common questions about dosage, ingredients, stock, delivery, returns, and order status.
- No scalable self-service assistant is available on the website.
- Manual responses create inconsistent brand tone and product accuracy.
- Health supplement communication has compliance risk if benefits are overstated.
- Leadership lacks visibility into query patterns, peak hours, and unresolved issues.
- Escalations are not consistently tagged by urgency, category, or customer intent.

Business Impact

Area	Current Risk	Desired Outcome
Support workload	Agents spend time on repeated questions.	40-60% routine question deflection.
Customer experience	Delayed first replies and inconsistent answers.	Instant first response for common questions.
Compliance	Risk of inaccurate supplement claims.	Approved, constrained answers with escalation.
Operations	Limited visibility into volume and topic trends.	Dashboard for query, category, order, and inventory signals.

3. Proposed Solution

Build a Health360 AI support assistant embedded on the Health360 website. The assistant answers approved product and policy questions, recommends relevant supplements based on user goals, handles order-support prompts, and escalates uncertain or sensitive issues to a human support process. The system includes a storefront experience, AI chatbot widget, full-screen chatbot testing mode, and admin dashboard for products, customers, and orders.

Solution Components

- Customer-facing chatbot: Website widget for product, order, shipping, return, and support questions.
- Knowledge-based responses: Uses Health360 catalog, inventory, policies, and business information.
- Admin dashboard: Tracks products, customers, orders, low stock, revenue, and support-related signals.
- Escalation layer: Routes low-confidence, complaint, medical, or human-agent requests for manual handling.
- Branded product catalog: Health360 supplement images and browsable product cards.

Solution Principles

- Give fast, concise, helpful responses for routine queries.
- Use Health360-approved product and policy data as the source of truth.
- Avoid diagnosis, treatment promises, or unsupported medical claims.
- Escalate when the query is uncertain, risky, emotional, or outside approved content.
- Make the system measurable through deflection, escalation, response-time, and CSAT metrics.

4. Goals and KPIs

ID	Goal	Target
G-01	Reduce repetitive tickets reaching agents.	40-60% deflection within 3 months.
G-02	Improve first response time.	Bot acknowledgement under 1 second; AI response target under 3 seconds.
G-03	Maintain safe supplement communication.	Zero known regulatory violations from bot content.
G-04	Increase customer satisfaction.	CSAT at or above 4.2 out of 5.
G-05	Improve operational visibility.	Dashboard for volume, products, customers, orders, revenue, and low stock.
G-06	Lower human support load.	At least 40% reduction in repetitive handling effort.

5. Scope

In Scope

- Website chatbot widget and full-screen chatbot testing experience.
- Health360 product catalog browsing, filtering, and branded product card experience.
- Admin product, customer, and order management pages.
- AI responses grounded in product catalog, category, inventory, business policies, and support guidance.
- Basic order support flow, including tracking guidance and human escalation.

- Query classification by broad intent: product query, order support, complaint, general support, medical/risk.
- Escalation guidance when confidence is low or user requests a human agent.
- Dashboard metrics for products, customers, orders, revenue, and low stock.

Out of Scope for Phase 1

- WhatsApp Business API automation.
- Email parsing and auto-response workflows.
- Voice bot, IVR, or call center integration.
- CRM, ERP, logistics, payment gateway, or warehouse integrations.
- Mobile app integration.
- Multilingual support beyond English.
- Automated refund approval or medical advice.

6. Users and Stakeholders

User / Stakeholder	Needs	Product Support
Customer	Fast answers about supplements, delivery, returns, and orders.	Website chatbot and product catalog.
Support Agent	Fewer repetitive tickets and clearer escalations.	Deflection, category tagging, escalation prompts.
Support Lead	Visibility into query trends, volume, and unresolved issues.	Admin dashboard and future reports.
Compliance Officer	Safe, approved supplement claims.	Guardrails, restricted response policy, escalation.
Operations / Inventory	Awareness of product and stock patterns.	Inventory display and low-stock dashboard.
Leadership	Lower cost, better service, measurable ROI.	Deflection and performance KPIs.

7. Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-01	Website chatbot widget	Critical	User can open chat, type, send, and receive a response.
FR-02	Full-screen chatbot	High	User can open full-screen chat and close it with UI or Escape.
FR-03	Product-aware answers	Critical	Bot answers catalog questions using product, price, category, and inventory data.
FR-04	Product recommendations	High	Bot recommends relevant supplements without medical claims.
FR-05	Order support guidance	High	Bot asks for order ID/email and routes complex cases to support.

FR-06	Query classification	High	Messages are classified as product, complaint, order, general, or medical/risk.
FR-07	Complaint escalation	Critical	Bot escalates complaints, refund disputes, legal threats, and safety concerns.
FR-08	Medical guardrails	Critical	Bot avoids diagnosis, treatment, disease-cure claims, and unsafe dosage advice.
FR-09	Admin dashboard	High	Dashboard renders products, customers, orders, revenue, and low-stock metrics.
FR-10	Product management	High	Admins can create, edit, search, delete, and export products.
FR-11	Customer management	Medium	Admins can create, edit, search, delete, and export customers.
FR-12	Order management	High	Admins can view order details and update order status.
FR-13	Storefront catalog	High	Users can browse product cards with brand imagery, price, rating, description, and SKU.
FR-14	Catalog filters	Medium	Users can filter by search, category, form, and sort option.
FR-15	Cart	Medium	Users can add, remove, increment, decrement, and clear cart items.
FR-16	Fallback data	High	Storefront and admin pages remain usable if Supabase is unavailable.
FR-17	Reporting export	Medium	Admin tables can export current rows to CSV.
FR-18	Brand presentation	Medium	Category cards show Health360-branded supplement imagery.

8. Non-Functional Requirements

ID	Category	Requirement	Target
NFR-01	Availability	Customer-facing chatbot and storefront should be available during business-critical hours.	99.5% monthly uptime target after production launch.
NFR-02	Performance	Chatbot responses should feel instant.	UI acknowledgement under 1 second; AI response target under 3 seconds.

NFR-03	Scalability	System should support growth without redesign.	Support up to 10,000 queries/day with horizontal scaling.
NFR-04	Security	API keys and privileged database operations must not be exposed to browsers.	AI and admin-sensitive operations run through server-side endpoints or protected roles.
NFR-05	Privacy	Customer and order information must be protected.	Collect only required data and restrict admin access.
NFR-06	Compliance	Bot must avoid prohibited health, disease, and cure claims.	All health content follows approved claim library and escalation policy.
NFR-07	Reliability	System should degrade gracefully when third-party services fail.	Fallback responses and seed data keep demo/storefront usable.
NFR-08	Observability	Support and engineering need visibility into failures and journeys.	Log chatbot failures, escalations, latency, and admin CRUD failures.
NFR-09	Auditability	Conversation outcomes should be reviewable for training and compliance.	Store metadata, category, escalation status, and approved response version.
NFR-10	Maintainability	Code should be modular and easy to extend.	Chat context, AI client, UI, data access, and visuals remain separated.
NFR-11	Accessibility	Core UI should work with keyboard and screen readers.	Accessible labels, keyboard dialogs, and WCAG AA contrast where practical.
NFR-12	Usability	Customers should understand how to use chat and catalog quickly.	Visible chat widget, quick actions, scannable cards.
NFR-13	Data Integrity	Admin edits must preserve valid data.	Required fields, numeric validation, valid status values, unique SKU/email constraints.
NFR-14	Backup and Recovery	Operational data should be restorable.	Supabase backup/export process documented before production launch.
NFR-15	Compatibility	App should work across modern browsers and responsive layouts.	Latest Chrome, Safari, Edge, and mobile responsive viewports.
NFR-16	Cost Control	AI usage should remain financially predictable.	Track usage, define max tokens, add fallback paths, and monitor provider cost.

9. Data and System Requirements

Entity	Purpose	Important Fields
Product	Catalog and recommendation source.	Name, SKU, category, form, brand, price, inventory, status, description.

Customer	Support and order ownership.	Name, email, phone, city, country, created date.
Order	Order support and admin workflow.	Customer ID, status, subtotal, tax, total, created date.
Order Item	Line-item detail for order view.	Order ID, product ID, quantity, unit price.
Conversation	Future production audit and analytics.	User message, bot response, category, confidence, escalation flag, timestamp.

System Dependencies

- Frontend: Next.js, React, TypeScript, Tailwind CSS.
- Database: Supabase for products, customers, orders, and order items.
- AI Provider: OpenRouter-backed assistant in the current prototype; production should route through server-side APIs.
- State: Zustand local cart persistence for storefront cart.

10. Compliance and Safety Rules

- Bot must not diagnose diseases or recommend treatment plans.
- Bot must not claim a supplement cures, prevents, or treats a disease unless specifically approved by compliance.
- Bot must recommend consulting a healthcare professional for pregnancy, chronic disease, medication interactions, severe symptoms, or dosage uncertainty.
- Bot must not replace doctor, pharmacist, or nutritionist guidance.
- Bot must not invent ingredients, certifications, clinical proof, delivery promises, or refund approvals.
- Every risky query should be escalated or answered with conservative general information.

11. Release Plan

Phase	Timeline	Activities	Exit Criteria
Discovery	Week 1	Finalize FAQs, product data, policies, escalation rules, compliance restrictions.	Approved knowledge base and escalation matrix.
Prototype	Week 2	Configure storefront, admin dashboard, chatbot context, and demo data.	Internal demo answers core product/order questions.
Compliance Testing	Week 3	Test risky prompts, medical questions, complaints, and edge cases.	Compliance signs off on guardrail behavior.
Pilot	Week 4	Launch to limited website traffic or internal users.	Bot resolves routine questions with acceptable CSAT and escalation accuracy.
Production	Week 5+	Full website rollout, monitoring, reporting, and optimization.	KPIs tracked weekly and improvement backlog created.

12. Risks and Mitigations

Risk	Impact	Mitigation
Bot provides inaccurate supplement guidance.	High compliance and customer safety risk.	Use approved knowledge base, guardrails, and human escalation.
API key exposure in browser.	High security and cost risk.	Move AI calls to server-side route before production.
Overly permissive database access.	High data integrity and privacy risk.	Use authenticated admin access and restrictive RLS policies.
Customers distrust bot responses.	Medium adoption risk.	Use clear brand tone, concise answers, and easy human handoff.
Low-quality knowledge base.	High answer quality risk.	Maintain product, policy, and FAQ data through support/product owner.
Provider latency or downtime.	Medium service risk.	Add fallback responses, retries, and status monitoring.

13. Open Questions

- Which exact FSSAI/UK ASA approved claims can be used for each product category?
- What CRM, ticketing, email, or WhatsApp platform should escalations integrate with in Phase 2?
- What are the official support hours for India and UK customers?
- Which analytics events should be included in the weekly leadership report?
- What customer data retention policy should apply to chat transcripts?

14. Acceptance Summary

The Phase 1 product is acceptable when customers can use the website chatbot to receive safe product and order-support guidance, admins can manage core catalog/customer/order data, risky queries are escalated rather than over-answered, and leadership can evaluate deflection and support workload improvement through measurable KPIs.